

THE PENNY MAGAZINE



Issue 35

October 20, 1832

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OF THE

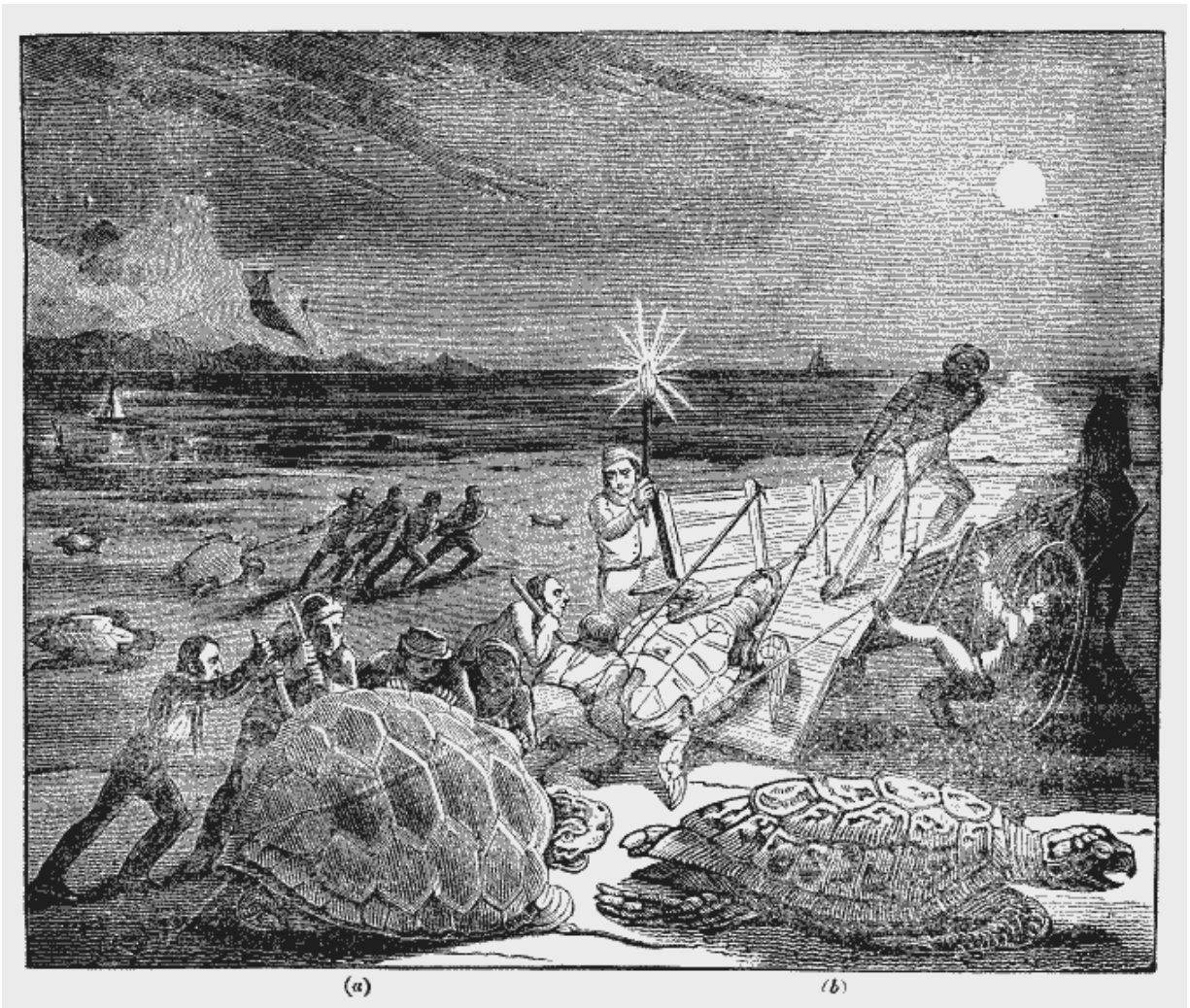
Society for the Diffusion of Useful Knowledge.

35.]

PUBLISHED EVERY SATURDAY.

[OCTOBER 20,
1832

CATCHING TURTLES.



[Catching Turtles on the Coast of Cuba.]

It is not improbable that some of our readers, who reside near a great commercial port, may have seen the landing of a cargo of strange-looking animals, which, turned upon their backs, appear the most helpless of creatures, and in this condition may have naturally led the spectator to imagine that they are incapable of removing from place to place, and have therefore little enjoyment of existence. These creatures, to use the language of the epicure, are fine “lively turtles”—the term “lively” being understood to mean that they have suffered little from a long voyage—that they are in good health—and that the “green fat,” the glory of aldermen, is in the most perfect state of excellence. Without asking our readers to feel any very strong interest in the prospects of high living which the arrival of a cargo of

turtles offers to many individuals who are somewhat too much inclined to set a high value upon the gratifications of the palate, we may be able to satisfy a rational curiosity as to the habits of these singular animals, which offer some higher benefits to mankind than that of furnishing the most costly luxury of a city feast.

The turtle and the tortoise belong to the same group of reptiles—in fact the turtle is a tortoise which principally inhabits the water, and is only found occasionally on the land. The two varieties represented in the above plate are the Green Tortoise (*a*), and the Loggerhead Tortoise (*b*). The former is the species chiefly used for food. It is found, in great numbers, on the coasts of all the islands and continents of the torrid zone. The shoals which surround these coasts are covered with marine plants; and in these water pastures, which are near enough to the surface to be readily seen by the naked eye in calm weather, a prodigious abundance of animals, mostly amphibious, feed, and amongst them multitudes of tortoises. Dampier, the old voyager, describing the Gallapagos Islands, says, “There are good wide channels between these islands fit for ships to pass; and in some places shoal water, where there grows plenty of turtle grass; therefore these islands are plentifully stored with sea turtle.” The tortoise, whether of the land or water species, is, as most of our readers know, protected, both on the back and belly, by a hollow shield, which is open at each end, for the issuing of the head and fore-feet at one time, and the tail and hind-feet at another.

The upper shield is termed the back-plate, or buckler; the lower shield the breast-plate. The middle of the buckler, in most of the species, is covered by numerous pieces or plates resembling horn in texture and composition; and the beautiful substance known by the name of tortoise-shell is obtained principally from a small species called the Hawksbill. The feet of the marine tortoises are much longer than those of the land, and their toes are united by a membrane so that they swim with great facility. The head, feet, and tail are covered with small scales. The jaws of the wide mouth are not provided with teeth, but the jaw-bones are very hard and strong, and being at the same time very rough, the animal is enabled to consume its vegetable food with ease, and at the same time to crush the shell-fish on which the marine species also feed. The green tortoise attains an enormous size and weight; some individuals measuring six or seven feet in length from the tip of the nose to the extremity of the tail, by three or four feet broad, and weighing as much as eight hundred pounds. Dampier says, “I heard of a

monstrous green turtle once taken at Port Royal, in the bay of Campeachy, that was four feet deep from the back to the belly, and the belly six feet broad. Captain Rocky's son, of about nine or ten years of age, went in it (meaning in the shell) as in a boat, on board his father's ship about a quarter of a mile from the shore." The green tortoise commonly weighs from two to three hundred pounds.

The instinct which leads the female turtle to the shore to lay her eggs, exposes her to the danger of becoming the prey of man. She deposits her eggs on the loose sand, and abandons them at once to the chance, which approaches almost to a certainty in the southern hemisphere, that they will be hatched by the influence of the sun's rays. She digs, by means of her fore-feet, one or more holes about a foot wide and two feet deep, in which she usually deposits more than a hundred eggs. These eggs are round, and are two or three inches in diameter; they are covered with a membrane something like wet parchment. The female generally lays three times in each year, at intervals of about a fortnight or three weeks. They almost always go ashore in the night time. A loose sand being essential to the hatching of the eggs, the turtles frequent only particular shores; but these are often several hundred miles from their feeding places. The eggs are hatched in less than a month after they are laid; and in about eight or ten days the young reptiles crawl to the water. Few, however, reach their native element, in proportion to the number produced. They become the prey of sea-fowl and various quadrupeds of prey. The tiger is an especial enemy to the tortoise; but man is still more actively engaged in their destruction. The collection of tortoise eggs forms one of the most important of the occupations of the Indians of the Orinoco. Humboldt has given a most interesting account of this branch of commerce, of which we shall furnish an abstract in a future number.

The wood-cut at the head of this article represents the manner in which the marine tortoises are caught on the coast of Cuba, and on parts of the South American continent. The Count de Lacepede, in his *History of Oviparous Quadrupeds*, has described the various modes in which the business of tortoise-catching is carried on; and we shall conclude this notice with an abstract of his account. It must be remarked that the turtle is a most important addition to the ordinary mode of victualling a ship; and that, therefore, the war in which the human race engages against them is rendered absolutely necessary by the wants of navigators. The turtles which

are demanded in England for the gratification of a luxurious appetite, constitute a very small number, when compared with those which offer an agreeable and salutary food to the hardy crews who are engaged in the commerce of the tropical seas.

“In spite of the darkness which is chosen by the female tortoises for concealment when employed in laying their eggs, they cannot effectually escape from the pursuit of their enemies: the fishers wait for them on the shore, at the beginning of the night, especially when it is moonlight, and, either as they come from the sea, or as they return after laying their eggs, they either dispatch them with blows of a club, or turn them quickly over on their backs, not giving them time either to defend themselves, or to blind their assailants, by throwing up the sand with their fins. When very large, it requires the efforts of several men to turn them over, and they must often employ the assistance of handspikes or levers for that purpose. The buckler of this species is so flat as to render it impossible for the animal to recover the recumbent posture, when it is once turned on its back.

“A small number of fishers may turn over forty or fifty tortoises, full of eggs, in less than three hours. During the day, they are employed in securing those which they had caught in the preceding night. They cut them up, and salt the flesh and the eggs. Sometimes they may extract above thirty pints of a yellow or greenish oil from one large individual; this is employed for burning, or, when fresh, is used with different kinds of food. Sometimes they drag the tortoises they have caught, on their backs, to enclosures, in which they are reserved for occasional use.

“The tortoise fishers from the West Indies and the Bahamas, who catch these animals on the coasts of Cuba and its adjoining islands, particularly the Caymanas, usually complete their cargoes in six weeks or two months; they afterwards return to their own islands, with the salted turtle, which is used for food both by the whites and the negroes. This salt turtle is in as great request in the American colonies, as the salted cod of Newfoundland is in many parts of Europe; and the fishing is followed by all these colonists, particularly by the British, in small vessels, on various parts of the coast of Spanish America, and the neighbouring desert islands.

“The green tortoise is likewise often caught at sea in calm weather, and in moon-light nights. For this purpose two men go together in a small boat, which is rowed by one of them, while the other is provided with a harpoon,

similar to that used for killing whales. Whenever they discover a large tortoise, by the froth which it occasions on the water in rising to the surface, they hasten to the spot as quickly as possible, to prevent it from escaping. The harpooner immediately throws his harpoon with sufficient force to penetrate through the buckler to the flesh; the tortoise instantly dives, and the fisher gives out a line, which is fixed to the harpoon, and, when the tortoise is spent with loss of blood, it is hauled into the boat or on shore.”

THE FLEMISH LANGUAGE.—No. 2.

Perhaps our readers may not be unwilling to see a few more specimens of the Flemish language. It should be stated that this book of dialogues, from which our last specimens were taken, contains at the end a kind of manual of good manners, it being the opinion of the writer—“that youth have long been in want of a treatise on manners, which should be based upon our usages, and adapted to the state of our knowledge.”

We give a few of the maxims of this Dutch Chesterfield for the use of those whom they may concern.

Directions for behaviour at table:—

Het <i>zoud</i> neémt men met het <i>punt</i> van het mes, het welk men aen zyn <i>brood</i> moet afvaegen.	You must take the <i>salt</i> with the <i>point</i> of your knife, after having wiped it on your <i>bread</i> .
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Instead of a literal translation we have given the meaning, marking with Italics a few words which are the most striking in the two languages. But besides those which we have marked, other similarities, such as *welk*, “whilk or which,” will be detected in this and the following specimen:—

<i>Daer is niets zoo</i> <i>ongemanierd</i> dan <i>iet</i> van de schotel te neémen	<i>There is nought so</i> <i>unmannered</i> as to take <i>ought</i> from the dish with the <i>fork</i> wherewith you eat,
---	--

met het *forket* waer men
mede eet.

or, literally, “*where man*
with eats.”

We think so; and we also say with the Dutch Chesterfield—

<i>Opent den mond niet te wyt</i>	<i>Open not your mouth too</i>
<i>als gy eet.</i>	<i>wide when ye eat.</i>

The reason for this precept is obvious; the Dutch Chesterfield has, however, thought it necessary to tell his countrymen the *why* and *wherefore* of it.

<i>Vraegt niet te drinken</i>	<i>Ask not (for something) to</i>
<i>terwyl gy zup eet; wagt</i>	<i>drink the while ye eat</i>
<i>tot dat gy iet anders</i>	<i>soup; wait till that ye</i>
<i>geëeten hebt.</i>	<i>somewhat (ought) else</i>
	<i>have eaten.</i>

And, above all,—

<i>Gaept niet rond ter wyl gy</i>	<i>Look not (gape) round the</i>
<i>drinkt.</i>	<i>while ye drink.</i>

Perhaps these maxims may be enough for one lesson. The original goes on for some length, and, finally, admonishes young folks not to carry off cakes, apples, &c. in their pockets from the table. We cordially concur in this advice.

We shall conclude with directions for sitting at table:—

<i>Als gy zult gezeten zyn,</i>	<i>When ye shall be seated,</i>
<i>struht u niet uyt op uwen</i>	<i>stretch 'o not out upon</i>
<i>stoel; zet uwe beenen</i>	<i>your seat (stool); set o'</i>
<i>niet overeen; wiegt niet</i>	<i>legs (bones) not over</i>
<i>met uwen stoel; maer</i>	<i>(across) one another; rock</i>
<i>houd u regt, de voeten</i>	<i>not with 'o stool; but hold</i>
<i>op den grond geplaetst.</i>	<i>'o right (straight) the feet</i>
	<i>placed on the ground.</i>

Directions how to hold hats and reticules in company:—

<i>Eenen jongeling houd</i>	<i>A younker holds his hat</i>
<i>zynon hoed op zyne</i>	<i>(hood) upon his knees, but</i>
<i>knieën, zonder het</i>	<i>without letting you see the</i>

binnenste te *laeten zien*,
Eene *jonge dogter* houd
insgelyks *haere* tassel
op haere knieen.

inside (het binnenste.) A
young girl (daughter) *holds*
in like manner *her* bag on
her *knees*.

COMETS.—No. 1.

Most of our readers must have heard of the comet of Biela, which appears in the present year, and has caused no small alarm among those who are entirely ignorant of the nature of comets in general, and of the track of this particular one. We have met with an amusing little book on this comet of 1832, by Littrow, professor of astronomy at Vienna, from which we shall give the substance of a few extracts, that may not be uninteresting.

There are only four comets whose orbits are yet accurately known. That which appears in the present year is called Biela's comet, from its having been discovered by an Austrian officer of that name in Bohemia in 1826. Its period of revolution round the sun is six years and two hundred and seventy days. Though it had been seen before in 1772 and 1805, it was not known to be a comet of so short a period. In the present year, 1832, we shall have its fourth visit. On the 27th of next November the comet will be nearest to the sun, but even then about seventy-two millions of geographical miles distant from that body: and on the 22d of this month (October) it will be nearest to the earth, and at the distance of about forty-four millions of miles from us.

The number of comets must be very great, for the appearance of near five hundred has been recorded; and if we consider how many must have passed unnoticed in the early history of the world for want of persons to observe them, we may form some idea of the prodigious quantity of these bodies. From 1769 to 1807 no comet appeared that attracted any attention from people in general, though astronomers during this period observed no fewer than thirty-six. There being then so many of these wanderers whose course is unknown, it may be supposed a possible thing that one of them should run foul of the earth; and supposing it to be a body of any considerable

magnitude and density, such a shock would entirely put an end to the present order of existence. Setting aside however, the question as to the magnitude and real nature of comets, let us consider what chance there is of our knocking against the comet of the present year, which, from the position of its orbit, looks much more threatening than any other that is known.

On the 29th of the present month this comet of Biela will be distant from a certain point in the earth's *orbit* only about $2\frac{1}{3}$ of the earth's diameters, or about twenty thousand miles in round numbers. If the earth were at this very point of its annual track on the same 29th of October, it *might* happen that we should feel such effects from the comet, or from the enormous mass of vapour composing it (computed to be more than one hundred and fifty times greater than the mass of our earth,) as to destroy all animal and vegetable life. But as the earth will not be at this dangerous point till the 30th of November, or thirty-two days *later* than the comet, we shall have nothing to fear from it this time. For the earth moving in its orbit at the rate of about 67,680 geographical miles in one hour, it will be 51,978,240 miles distant from the comet on the 29th of this month, and in no danger at all of being affected by it in any way that we can estimate.

Perhaps few people will trouble themselves about this comet any more, when they learn that they are quite safe for the present. But how, it may be asked, are we sure that on some future occasion we may not approach too near? If the comet should be in its nearest point to the sun on the 28th of December, instead of the 27th of November, then we should really approach it within the short distance above-mentioned. But this near approach cannot take place, unless the comet should be in its nearest point to the sun in the latter part of December; and this again will not take place till the year 1933, when the comet will be in its perihelion (i. e. nearest point to the sun) on the 31st of December, and again in the year 2115, on the 26th of the same month. But, should the comet's period of six years and two hundred and seventy days be somewhat changed in the course of the next century, from the action of Jupiter and other planets, (which is far from improbable,) this would diminish still further the chance of any unpleasant proximity in the years 1933 or 2115. This we hope will console those who regard this visitor with more feelings of fear than curiosity.

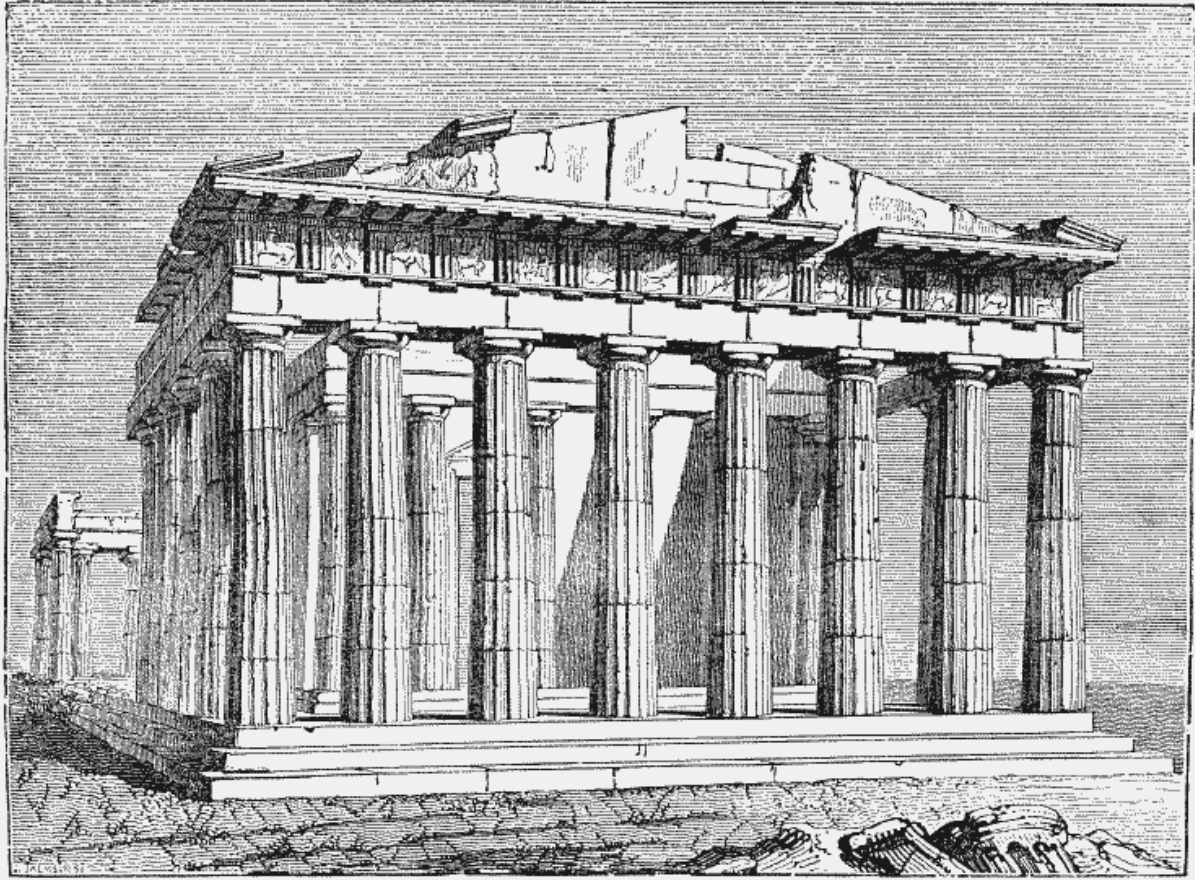
It may be added that this comet is a very small one, and, though its vapour occupies so enormous a space, the real kernel or bright part of the

comet is not more than sixty or eighty miles in diameter; and hence it is conjectured that if it really is a *body* properly so called, it must be very small indeed, and that, even in a near approach to the earth, any injury that it might do by its attraction would be hardly felt. Again, says Littrow—"as to the tail and its deadly vapours, which, as they say, threaten us with such dreadful consequences, we really have nothing at all to fear from *them*: and for the following plain, but quite satisfactory reason—the comet has no tail."

The following conclusion will, we hope, remove whatever apprehension may still lurk in the minds of the most timid, as to the danger which they have to fear from this comet in the years 1933, 2115, and in subsequent years—should their lives be so far prolonged.

"We have already stated that Biela's comet can only come near the earth when it is at its least distance from the sun, in the latter part of December. But since this proximity of the comet to the sun may just as well happen on *every other* day of the year as in December, and since its period is six years two hundred and seventy days, or about two thousand five hundred days, in round numbers,—after a lapse of two thousand five hundred years, a near *approach* (not an actual collision) to the comet is *probable*. I say merely *probable*, from which it must not be concluded that such an event actually will take place in two thousand five hundred years. This result merely means that a man might bet two thousand five hundred to ten, or to one hundred, that the comet will not come *near* the earth for the next ten or one hundred years. At the end of two thousand five hundred years there will be an equal chance that the comet will make this near approach, or that it will not. And after two thousand five hundred years the chance of its approaching the earth will go on increasing, but at so slow a rate that many thousands of years must elapse before the event can be *really* expected."

THE PARTHENON.



[Remains of the Parthenon.]

We shall proceed with our description of the Athenian antiquities in the British Museum, as soon as the collection is numbered according to its present arrangement. We understand from the Officers of the Institution that this essential assistance to the visitor will be immediately given; for the old order of the several pieces of sculpture being considerably altered, a reference to the former numbers only would prove embarrassing. In the mean time we give that view of the Parthenon, for which the representation of the temple of Apollo Epicurius, near Phigalia, was substituted by mistake.

NAPLES.

In a preceding number we endeavoured to give our readers a notion of the situation and main features of Naples. We shall now devote a page to a few of the interesting objects contained within that city.

The first in importance is the *Studj* or *Museo Borbonico*, or, what we may better call, the National Museum. In many respects this magnificent establishment is unrivalled in the world. Besides a rich statue gallery which boasts the Farnesian Hercules, the all perfect Aristides, the Farnesian Toro, a Venus perhaps superior in loveliness to the Medicean, and other masterpieces of ancient Greek art, the Museum contains a gallery of pictures with two of Raphael's best works, and splendid specimens of Titian, Correggio, Claude, Salvator Rosa, and other great masters; and, moreover, a library, a collection of Etruscan vases, a cabinet of ancient coins and medals, and rooms filled with the ancient relics of Herculaneum and Pompeii.

The collection of vases, which have nearly all been discovered and dug up in the kingdom, is the richest in existence; but it is more especially the collection of the objects rescued from the two interred cities, that gives the Museum of Naples its superiority to others.

In this collection are found some of the most perfect works of ancient art in bronze, domestic implements of nearly every sort, mechanical tools, surgical and mathematical instruments, rings, necklaces, and other specimens of jewellery, and even the entire apparatus of a woman's toilet. The attentive visitor, by studying these objects, may in a few hours obtain a better insight into the domestic manners of the ancients, than whole years devoted to books can give him. One of the most interesting departments of this unique collection, is that of the papyri, or manuscripts, discovered in the excavations of Herculaneum. The ancients did not bind their books (which, of course, were all manuscripts) like us, but rolled them up in scrolls. When these of Herculaneum were discovered, they presented, as they still do, the appearance of burnt sticks, or cylindrical pieces of charcoal, which they had acquired from the action of the heat contained in the lava that buried the whole city. They seem quite solid both to the eye and touch, yet an ingenious monk discovered a process of detaching leaf

from leaf and unrolling them, by which they could be read without much difficulty. When these manuscripts were first exposed to the air a considerable number of them crumbled to dust. Our countryman, the late Sir Humphry Davy, destroyed the integrity of a few by making unsuccessful experiments, which he fancied might produce a result that would supersede the slow and laborious process now adopted; but about eighteen hundred still remain. Four of them have been unrolled, and fac-similies of them, with translations, published by the Neapolitan government.

To pass to a very different object. One of the singularities of Naples is its Campo Santo, or cemetery for the poor. This is situated on the skirts of the town, looking towards Mount Vesuvius. A wall of inconsiderable elevation encloses a quadrangular space, whose surface is cut into three hundred and sixty-five holes, like the mouths of wells or cisterns. One of these holes is opened every day; the dead bodies of the poor of that day—without coffins—without so much as a rag about them—are thrown one upon another, as they arrive, through the mouth into a deep cave below cut in the tufa rock, and at night a stone is laid over the horrid sepulchre and secured by cement. The next day the cave next in order of date is opened, and so on through the year. At the end of the year, the first cave is again opened, by which time its contents, the decomposition of which is assisted by quick-lime, are reduced to little more than bones.

The catacombs of Naples, whose entrance is under the hill of Capo-di-Monte, and the grotto of Posilippo, at the extremity of the western suburb of the city, are also remarkable objects. The first are of great extent, and contain many curious specimens of painting and subterranean architecture by the early Christians, and an appalling mass of human skulls and bones, the relics of the victims of a plague that depopulated Naples some two centuries back. The second is a subterranean passage cut through the hill of Posilippo in remote antiquity, but enlarged and improved as a road in modern times. It is considerably more than half a mile long by twenty-four feet broad; its height is unequal, varying from twenty-five to sixty feet: it is well paved with large flags of lava. By night, it is *now* tolerably well illuminated by lamps suspended from its rugged roof, but by day the “darkness visible” that reigns through the passage renders it always solemn and sometimes embarrassing. Being the only frequented road to and from the town of Pozzuoli, Baia, Cuma, and other places, there is seldom a lack of passengers; and their voices, as they cry to each other in the dark, and the

noise of their horses' tread and of the wheels of their waggons, carriages, and gigs, echoing through the grotto and the deep vaults which in many places branch off from it laterally, produce to the ear of the stranger an effect that is almost terrific. Immediately above the entrance to the grotto, coming from the city, stands on a romantic cliff, which has been in part cut away to widen the approach to the subterranean road, an ancient Roman tomb in almost perfect preservation. This tomb is supposed to have been that of the great poet Virgil, and is visited as such by every traveller. Its claim has been questioned in vain; mankind are attached to such pleasant illusions, (if this be one, which we by no means decide,) and continue from age to age to crowd to the spot. A laurel once flourished by the side of the venerable sepulchre and covered its roof; but the successive thousands and thousands of visitors, each anxious for a memorial gathered in such a spot, have not left leaf, branch, stem, or root of the sacred tree.

In the old part of the city, among some Roman ruins called the "*Anticaglia*," are supposed to exist part of the walls of the theatre where the Emperor Nero sang and played on the lyre like a common actor. The Neapolitans care little about this; but their great boast, that which they fancy renders them the envy of the world, is their Opera-house of San Carlo, which in truth, must be acknowledged as the most spacious and most splendid theatre in Europe.



FRACTIONS.

It is not our intention to write a treatise on the part of arithmetic which stands at the head of this article, or to enter into the reasons why so many persons, who can solve a simple question in which there are nothing but whole numbers, are puzzled by anything which contains fractions. Our object is, to give some slight notions on this part of the subject to those who are already able to work the four rules in whole numbers.

When we add any two numbers together, it is understood that both of them have the same unit, or that both are some number of times the same thing. Thus, that two and three make five, means that two *yards* and three *yards* make five *yards*, or that two *pounds* and three *pounds* make five *pounds*, and so on. We do not in that case say anything of two *yards* and three *feet*, or of two *pounds* and three *shillings*. The following questions might arise:—If we have a distance which is neither six yards nor seven yards, but something between the two, how are we to represent this in numbers, and form rules for adding and subtracting this length to or from others of the same kind, without introducing a new measure, or talking of any other length except a yard? The answer to this will bring us, as we shall see, to the common meaning of the word *fraction*, and the way of representing a fraction. As we cannot measure anything exactly, we must first decide what degree of accuracy is necessary. This will vary in different operations, but we will suppose, for example's sake, that a line may be rejected as insignificant, of which it would take more than a hundred to make a yard. If then we divide a yard into one hundred equal parts, and first remove the six whole yards which the above-mentioned distance contains, we have a remainder which does not contain all the hundred parts just mentioned, since it is less than one yard. Suppose that, on measuring the remainder, we find it to contain more than 53 and less than 54 of the hundred parts: if then we call it 53 parts out of a hundred of a yard, the error

committed will be less than one part out of a hundred; that is, by what was supposed above, it will be sufficient to say that the length of the whole is 6 yards and 53 of the hundred equal parts which would compose another yard, or 53 hundredths of a yard. If we were inventing a system of arithmetic, we might choose among many different ways of representing this. For example, 6 yards 53_{100} of a yard; 6 yards and $53 \div 100$ of a yard; and so on. The common method is the following, $6\frac{53}{100}$ yards, it being always understood that when we write two numbers under one another with a line between, the unit of which we speak, be it a yard, pound, acre, or any other, is cut into as many equal parts as are shown by the lower number, and as many of them are taken as is shown by the higher number. Thus, $\frac{7}{8}$ of a mile is the length obtained by cutting a mile into 8 equal parts, and taking 7 of them, being of course less than the whole mile by one of these parts.

Such a fraction as we have described is less than the unit of which it is a part; but a whole number of units and a fraction may be represented together by the same method. If, in the preceding example, we had divided each of the six yards into 100 parts, there would have been 600 such parts, which, with the 53 parts furnished by the fraction, would have made 653, not of yards, but of the hundredth parts of yards. This we should represent by $\frac{653}{100}$, denoting that each of a succession of yards has been divided into 100 parts, out of which collection of parts 653 have been taken. The term fraction is applied equally to all cases; and with this extension of meaning, the unit itself maybe represented as a fraction, for one yard is $\frac{2}{2}$ yards, or $\frac{3}{3}$ yards, or $\frac{4}{4}$ yards, and so on.

The lower line of a fraction is called the *denominator*, and the upper the *numerator*: these are Latin words, which may be literally translated by the *namer* and the *numberer*; the first tells *what sort* of parts is taken, and the second *how many* of them are taken. The following propositions will serve for consideration, and also to familiarize the reader with the use of these terms. When the numerator is less than the denominator, the fraction is less than a unit. When the numerator is greater than the denominator, the fraction is greater than the unit. Of two fractions which have the same denominator, that is the greater which has the greater numerator. Of two fractions which have the same numerator, that is the greater which has the less denominator. It is usual to distinguish fractions which are less than the

unit from those which are greater by calling the former *proper*, and the latter *improper*, fractions.

As yet we have only considered fractions of the unit; and it is always understood that a simple fraction, such as $\frac{7}{8}$, is a fraction of the unit, or it is *one* yard or *one* pound which is divided into 8 parts. Fractions of other numbers are written by placing the number to be divided after the fraction of it which is to be taken, thus— $\frac{3}{4}$ of 7, which means that 7 is to be divided into 4 parts, of which parts, 3 are taken. We now ask, what fraction of the unit is $\frac{3}{4}$ of 7, or into how many parts must *one* yard be cut, and how many times must one of those parts be repeated, so as to give the same length which arises from cutting *seven* yards into 4 parts, and taking 3 of them? It is obvious that $\frac{3}{4}$ of 7 yards is 7 times as much as $\frac{3}{4}$ of 1 yard, or simply $\frac{3}{4}$ and 3 quarters of a yard repeated 7 times is 21 quarters or $\frac{21}{4}$. Similarly $\frac{2}{5}$ of 8 is $\frac{16}{5}$ of 1, or $\frac{16}{5}$. Hence it follows that $\frac{1}{4}$ of 3 is $\frac{3}{4}$, $\frac{1}{9}$ of 13 is $\frac{13}{9}$, and so on. If therefore we take the eighth part of nine, we get the same as if we had repeated the eighth part of the unit nine times. We may therefore consider a fraction, such as $\frac{5}{6}$, in two ways, either as the sixth part of five, or as the sixth part of unity repeated 5 times. It may sometimes be necessary to take a fraction of a fraction, such as $\frac{2}{5}$ of $\frac{7}{8}$, or having found $\frac{7}{8}$ of 1, to divide it into five parts, and take two of them. We ask, what fraction of the unit would the result of this double operation give? The answer is multiply the two numerators together, and also the two denominators, which gives $\frac{14}{40}$, or two-fifths of seven-eighths of a yard is fourteen parts out of forty. To see the reason, let us first take the more simple case $\frac{1}{5}$ of $\frac{1}{8}$. It is plain that if we divide one yard into eight equal parts, and afterwards divide each of these parts into 5 equal parts, we have divided the whole yard into 8 times 5, or 40 equal parts. Consequently the fifth part of an eighth part is one fortieth of the whole, or $\frac{1}{5}$ of $\frac{1}{8}$ is $\frac{1}{40}$. But one fifth of *seven* eighths will be 7 times as much as $\frac{1}{5}$ of *one* eighth, and will therefore be $\frac{7}{40}$; again, two fifths of $\frac{7}{8}$ will be twice as much as *one* fifth of $\frac{7}{8}$, and will therefore be $\frac{14}{40}$, or $\frac{2}{5}$ of $\frac{7}{8}$ is $\frac{14}{40}$, according to the rule. In the same way $\frac{3}{7}$ of $\frac{11}{10}$ is $\frac{33}{70}$. This rule corresponds to the multiplication of whole numbers, and is therefore called multiplication of fractions. The connexion is not obvious at first, owing to a little difference in our manner of speaking about whole numbers and fractions. But if we were in the habit of saying that 2 multiplied by 6 is six *of* 2, in the same way as we say “six of them,” “six of

his men," it would appear natural to call those rules which tell us how many units there are in six *of* two, and what fraction of unit there is in $\frac{2}{5}$ *of* $\frac{7}{8}$, by the same name. By this rule all questions of fractions are solved, which would have required multiplication if they had been in whole numbers. For example, if 1 pound cost 2 shillings, 6 pounds will cost 6 times 2 shillings; similarly, if 1 pound costs $\frac{7}{8}$ of a shilling, $\frac{2}{5}$ of a pound will cost $\frac{2}{5}$ of $\frac{7}{8}$ of a shilling.

The most important proposition relating to fractions, being the one on which the rules most materially depend, is the following: If the numerator and denominator be either both multiplied or both divided by the same number, the value of the fraction is not altered. For example, take $\frac{3}{5}$ and multiply its numerator and denominator by 4, which gives $\frac{12}{20}$. In the second fraction we cut the unit into four times as many parts as in the first, consequently each part of the unit signified in the second fraction is the fourth of that signified in the first. But in the second fraction, four times as many parts are taken as in the first, by which the balance is restored. Let us suppose that two yards of cloth are to be measured by a foot measure. The foot being $\frac{1}{3}$ of the unit, and 6 of these being necessary, $\frac{6}{3}$ will be the fraction in yards, representing not only the number of yards measured, but in what parts of yards they were measured. No one would object to an inch measure, which is $\frac{1}{12}$ of a foot, provided 12 times as many inches were given as there were feet in the first case. But one inch is $\frac{1}{36}$ of a yard, and 12 times 6 is 72; and in this way of measuring $\frac{72}{36}$ would represent the number of yards given, which is derived from $\frac{6}{3}$ by multiplying the numerator and denominator by 12. Similarly, one shilling, the unit being a pound, is $\frac{1}{20}$; and 12 pence, the unit being also a pound, is $\frac{12}{240}$; and $\frac{1}{20}$ and $\frac{12}{240}$ only differ in that the numerator and denominator of the first must be multiplied by 12 in order to make the second.

Hence it is allowable to multiply the numerator and denominator of a fraction by any number which is convenient, and which is called multiplicand, since that operation does not alter its value. Thus, $\frac{2}{3}$, $\frac{4}{6}$, $\frac{6}{9}$, $\frac{8}{12}$, &c. are all of the same value, when the unit is the same in all: in common language, we should say, that two out of three is the same as four out of six, six out of nine, and so on. We are now able to remove two fractions which have different denominators, and substitute others of the

same value with the same denominator. Take the fractions $\frac{2}{3}$ and $\frac{4}{5}$. If we ask which is the greater, no answer can at first be given, for though the second, is 4 and the first 2, yet the second is four of the *fifth* parts only of unity, while the first is 2 of the *third* parts. But if we multiply the numerator and denominator of each fraction by the denominator of the other, the results will be $\frac{10}{15}$ and $\frac{12}{15}$, which have the same value $\frac{2}{3}$ and $\frac{4}{5}$, and also have the same denominator as each other. Hence we see that, $\frac{12}{15}$ being greater than $\frac{10}{15}$, $\frac{4}{5}$ is greater than $\frac{2}{3}$. The sum of the two is the fifteenth part of unity repeated 22 times or $\frac{22}{15}$; the difference is two parts out of fifteen or $\frac{2}{15}$. Hence follow the common rules for addition and subtraction of fractions.

We now come to the reverse of multiplication. We have shown how to find the value of one fraction of another, such as $\frac{3}{5}$ of $\frac{6}{11}$; we now ask, what fraction of $\frac{7}{8}$ must be taken, to give $\frac{2}{3}$ of 1 or simply $\frac{2}{3}$? Into how many parts must we cut $\frac{7}{8}$, and how many times must we repeat one of those parts, in order that the result may be the same as if we had cut unity into three parts, and taken 2 of them? Reduce the fractions $\frac{7}{8}$ and $\frac{2}{3}$ to other equivalent fractions having the same denominator, which are $\frac{21}{24}$ and $\frac{16}{24}$. If we cut $\frac{21}{24}$, which is $\frac{7}{8}$, into *twenty-one* equal parts, each of these parts is $\frac{1}{24}$; if we repeat $\frac{1}{24}$ *sixteen* times, the result is $\frac{16}{24}$, which is $\frac{2}{3}$: hence, if $\frac{7}{8}$ be cut into 21 equal parts and 16 of these parts be taken, the resulting fraction is $\frac{2}{3}$, or if we ask, what fraction of $\frac{7}{8}$ is $\frac{2}{3}$? the answer is $\frac{16}{21}$ of $\frac{7}{8}$. By our former rule $\frac{16}{21}$ of $\frac{7}{8}$ is $\frac{112}{168}$, which does not appear at first sight to be the same as $\frac{2}{3}$, but if we examine its terms, we shall find that on dividing the numerator and denominator by 56 (which does not alter its value) it is reduced to $\frac{2}{3}$. This rule being the reverse of multiplication is called division; the fraction which is to be cut into parts is called the *divisor*, that which is to be produced from it the *dividend*, and the fraction of the first, which it is necessary to take, in order to produce the second, is called the *quotient*. Thus, $\frac{16}{21}$ is the quotient of $\frac{2}{3}$ divided by $\frac{7}{8}$. The rule deduced from this reasoning is: Reverse the divisor, that is, for $\frac{7}{8}$ write $\frac{8}{7}$, and proceed as in multiplication with the reversed divisor and the dividend. Thus, $\frac{8}{7}$ of $\frac{2}{3}$ is $\frac{16}{21}$. This rule is used in every question where division would have been used, if whole numbers only had been given. Thus if 4 pounds cost 20 shillings, the price of one pound is found by dividing 20 by

4, and is 5 shillings. If $\frac{7}{8}$ of a pound cost $\frac{2}{3}$ of a shilling, the price of one pound is found by dividing $\frac{2}{3}$ by $\frac{7}{8}$ and is $\frac{16}{21}$ of a shilling. This might be established by independent reasoning as follows: As $\frac{7}{8}$ of a pound costs $\frac{2}{3}$ of a shilling, and 7 pounds cost 8 times as much as $\frac{7}{8}$ of a pound, 7 pounds will cost $\frac{16}{3}$ of a shilling. But as the price of one pound is one-seventh of that of 7 pounds, for every third of a shilling which 7 pounds cost, one pound will cost the twenty-first part of a shilling. Hence the price of one pound is $\frac{16}{21}$ as before.

We shall proceed in a future number to the explanation of Decimal Fractions.

THE WEEK.

October 27.—The birth-day of Captain Cook. James Cook was born in 1728 at the village of Marton in the North Riding of Yorkshire. His parents were of the class of labourers. All the education he received amounted only to English reading, writing and the elements of arithmetic. He was then, at the age of thirteen, bound apprentice to a small shopkeeper in the neighbouring town of Snaith, which is on the sea-coast. Here he became so smitten with the love of a sea-life that he could not rest till his wish was gratified; and his master was at last induced to let him off, when he entered himself as one of the crew of a vessel engaged in the coal trade. In this humble and laborious line of life he continued till the breaking out of the war of 1755. He then entered the navy, as a common seaman, of course. But now the native superiority of the man began to assert itself; and in four years he rose to be Master of the Mercury, one of the ships belonging to an expedition sent against Quebec. Thus by far the most formidable of the difficulties were overcome which he had to encounter in emerging from obscurity; he was now on the direct road to preferment, and in a position in which his good conduct and perseverance were sure to meet with their reward. While stationed in this command on the coast of North America, he greatly distinguished himself both by his skill and intrepidity as a seaman;

and he also made use of his leisure to rectify the defects of his original education by studying mathematics and astronomy. He eventually made himself in this way one of the most scientific naval officers of that time. His reputation rose accordingly; and in 1768, when Government resolved to send out the Endeavour to the South Sea to obtain an observation of the approaching transit of Venus, Cook was selected to command the ship. He conducted this expedition with admirable ability, and so entirely to the public satisfaction, that, having returned home in 1771, he was the following year appointed to proceed again to the same regions with two ships, the Resolution and the Adventure, with the object of endeavouring to settle the long-disputed question as to the existence of a southern polar continent. On this voyage, in which he circumnavigated the world, he was absent nearly three years; and notwithstanding all the vicissitudes of climate and weather, and the other dangers which he had encountered, he brought home, with the exception of one, every man of the crew he had taken out with him. He communicated to the Royal Society an account of the methods he had adopted on this occasion for preserving the health of his men; and that body in return elected him into their number, and voted him the Copley gold medal as a testimony of their sense of his merits. To crown his achievement, Captain Cook wrote the history of this expedition himself, and wrote it admirably. In little more than a year after his return, he sailed on his third and last voyage of discovery; the principal object of which was to ascertain the practicability of a passage between the Atlantic and Pacific Oceans along the northern coast of America. After having been out on this expedition nearly three years, and having explored a vast extent of sea and coast, the great circumnavigator put in at the island of Owhyhee on his return home; and he was there killed in a sudden and accidental rencontre with some of the natives on the 14th of February, 1779. The late Admiral Burney, who was present on this occasion, mentions, in a note to his History of Discoveries in the South Sea, an anecdote which deserves to be remembered. Of the party of marines, by whom Captain Cook was accompanied when he met his death, four were killed along with him; "and in the hasty retreat made," says Burney, "after the boats had put off, one man still remained on shore, who could not swim. His officer, Lieutenant (now Colonel) Molesworth Phillips, of the Marines, though himself wounded at the time, seeing his situation, jumped out of the boat, swam back to the shore, and brought him off safe." The author proceeds to

compare this conduct of Lieutenant Phillips with a similar act performed in 1624 by a Dutch Captain, Cornelys de Witte, who, when a boat's crew which he commanded was surprised in a port on the coast of America by an ambuscade of Spaniards, and driven to sea after four of them had been killed, seeing one of his men left behind on the beach, boldly returned to the shore in the face of the enemy, and took him into his boat. "This was an act of generosity," observes the French translator of the account of the Dutch voyage, "worth a wound which he received in his side, and of which he was afterwards cured." The news of the death of Cook was received by his countrymen, and it may be said by the world, with the feeling that one of the great men of the age was lost; and both in his own and in foreign nations public honours were liberally paid to his memory. In the half century of busy and enterprising exertion in every field of activity which has elapsed since his death, no newer name in the same department has yet eclipsed the lustre of his, and with reference to the peculiar character of his fame, as contrasted with that of our other renowned seamen, it has been well and justly remarked that, "while numberless have been our naval heroes who have sought and gained reputation at the cannon's mouth, and amidst the din of war, it has been the lot of Cook to derive celebrity from less imposing, but not less important exploits, as they tended to promote the intercourse of distant nations, and increase the stock of useful science."^[1]

¹. Gorton's Biographical Dictionary.



[Portrait of Captain Cook.]

DOMESTIC PEACE.

Tell me on what holy ground
May domestic peace be found?
Halcyon-daughter of the skies!
Far on fearful wings she flies
From the tyrant's scepter'd state,
From the rebel's noisy hate.
In a cottag'd vale she dwells,
List'ning to the sabbath bells,
While all around her steps are seen
Spotless Honour's meeker mien.
Love, the sire of pleasing fears,
Sorrow smiling through her tears;
And, mindful of the past employ,
Memory, bosom spring of joy!—COLERIDGE.

Ants of Brazil.—So numerous were the ants, and so great was the mischief which they committed, that the Portuguese called this insect the *King of Brazil*; but it is said by Piso, that an active husbandman easily drove them away, either by means of fire or of water; and the evil which they did was more than counterbalanced by the incessant war which they waged against all other vermin. In some parts of South America they march periodically in armies, such myriads together, that the sound of their coming over the fallen leaves may be heard at some distance. The inhabitants, knowing the season, are on the watch, and quit their houses, which these tremendous, but welcome visitors clear of centipedes, forty-legs, scorpion, snake, every living thing; and having done their work, proceed upon their way.—*Southey's Brazil.*

Singular Customs.—There is a custom, proper to Sicily, which I must not forget to mention. This is a right of purchase of a singular kind. If any man buy an estate, be it house, land, or vineyard, the neighbour of the purchaser, for the space of an entire year afterward, may eject him by an advance of price. In vain would the first purchaser give more to the original owner. This singular law is generally evaded by a falsehood. The purchase-money is stated, in the articles of agreement, at a higher sum than has been agreed upon in the presence of four witnesses. There is another no less singular law in Sicily, according to which any man can oblige his neighbour to sell his house, if he will pay him three times its value. The intention of this law was, the improvement of the towns. It was to encourage the possessors of large houses to purchase the humble abodes of the poor.—*Count Stolberg's Travels.*

Volcano in Iceland.—The Oræfa mountain is not only the loftiest in Iceland, but has been rendered remarkable by the great devastation made by its eruption about a century ago. Nothing can be more striking than the account of this calamity given by Jon Thorlakson, the aged minister of a neighbouring parish. He was in the midst of his service on the Sabbath, when the agitation of the earth gave warning that some alarming event was to follow. Rushing from the church, he saw a peak of the neighbouring mountain alternately heaved up and sinking; the next day, this portion of the mountain ran down into the plain, like melted metal from a crucible, filling it to such a height, that, as he says, no more of a mountain which formerly towered above it could be seen, than about the size of a bird; volumes of water being in the mean time thrown forth in a deluge from the crater, sweeping away whatever they encountered in their course. The Oræfa itself then broke forth, hurling large masses of ice to a great distance; fire burst out in every direction from its sides; the sky was darkened by the smoke and ashes, so that the day could hardly be distinguished from the night. This scene of horror continued for more than three days, during which the whole region was converted into utter desolation.—*North American Review for July, 1832.*

Farming in Iceland.—The most important branch of rural labour in Iceland, is the hay-making. About the middle of July, the peasant begins to cut down the grass of the *tûn* (the green around his house,) which is immediately gathered to a convenient place, in order to dry, and, after having been turned once or twice, is conveyed home on horseback to the yard, where it is made up into stacks. At the poorer farms, both men and women handle the scythe; but in general, the women only assist in making the hay after it is cut. In many parts of the island, where there is much hay, the peasants hire men from the fishing plains, who are paid for their labour at the rate of thirty pounds of butter a week. They cut by measurement; the daily task being about thirty square fathoms. Hay-harvest being over, the sheep and cattle that had been out all summer on the mountains are collected; the houses are put into a state of repair for the winter; the wood needed for domestic purposes is brought home to each farm; the turf is also taken in. During the winter, the care of the cattle and the sheep devolves entirely on the men; and consists chiefly in feeding and watering the former, which are kept in the house, while the latter are turned out in the day-time to seek their food through the snow. When the snow happens to be so deep that they cannot scrape it away themselves, the boys do it for them; and as the sustenance thus procured is exceedingly scanty, they generally get a little of the meadow hay about this time. The farm hay is given to the cows alone. All the horses, excepting perhaps a favourite riding horse, are left to shift for themselves the whole winter, during which season they never lie down, but rest themselves by standing in some place of shelter.—*Henderson's Iceland.*

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Transcriber's Notes

New original cover art included with this eBook is granted to the public domain. Itemized changes from the original text:

- [p. 282](#): Replaced “n” with “in” in phrase “the war in which the human race engages.”
- [p. 282](#): Replaced “endered” with “rendered” in phrase “is rendered absolutely necessary by the wants of navigators.”
- [p. 282](#): Replaced “s” with “as” in phrase “as they come from the sea.”
- [p. 283](#): Supplied partially misprinted words “which” and “actually” in phrase “from which it must not be concluded that such an event actually will take place.”
- [p. 284](#): Added closing double quotation mark after phrase “many thousands of years must elapse before the event can be *really* expected.”
- [p. 285](#): Changed “as we must first” with “we must first” in phrase “As we cannot measure anything exactly, we must first decide what degree of accuracy is necessary.”
- [p. 286](#): Replaced “ $\frac{4}{12}$ ” with “ $\frac{8}{12}$ ” in phrase “Thus, $\frac{2}{3}$, $\frac{4}{6}$, $\frac{6}{9}$, $\frac{8}{12}$, &c. are all of the same value.”
- [p. 287](#): Replaced “ast” with “last” in phrase “his master was at last induced to let him off.”
- [p. 287](#): Added comma after phrase “the Resolution and the Adventure.”
- [p. 287](#): Added comma after phrase “after four of them had been killed.”

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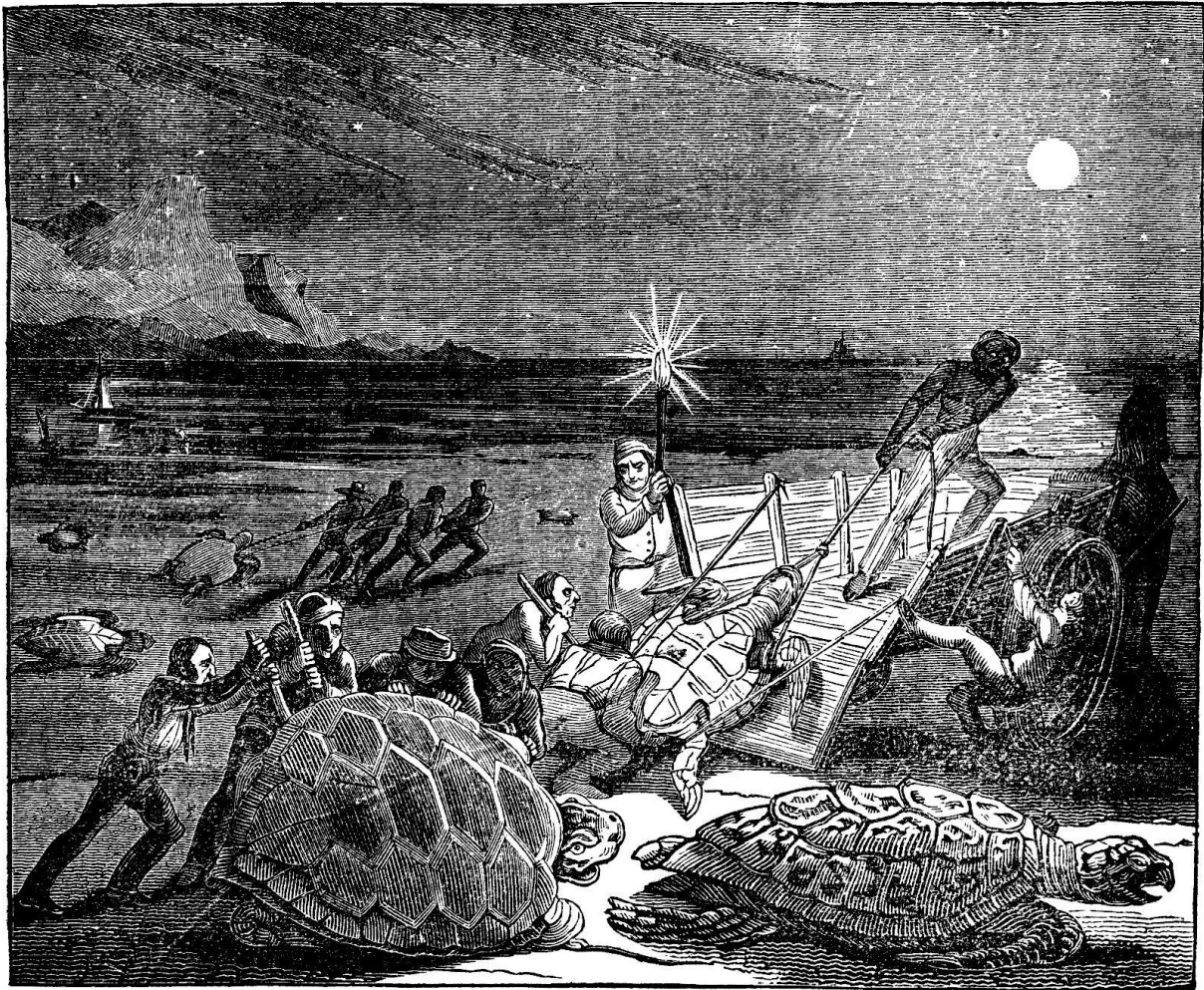
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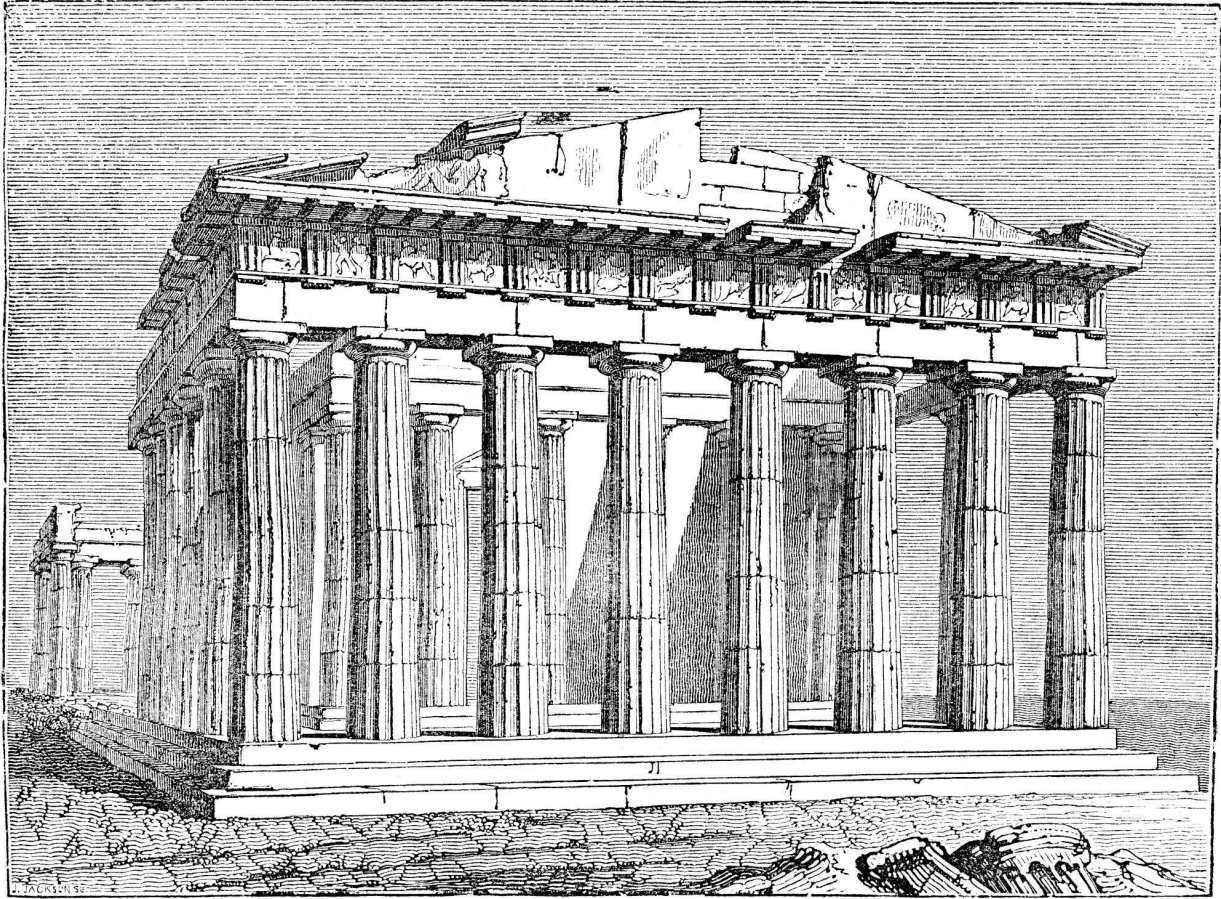
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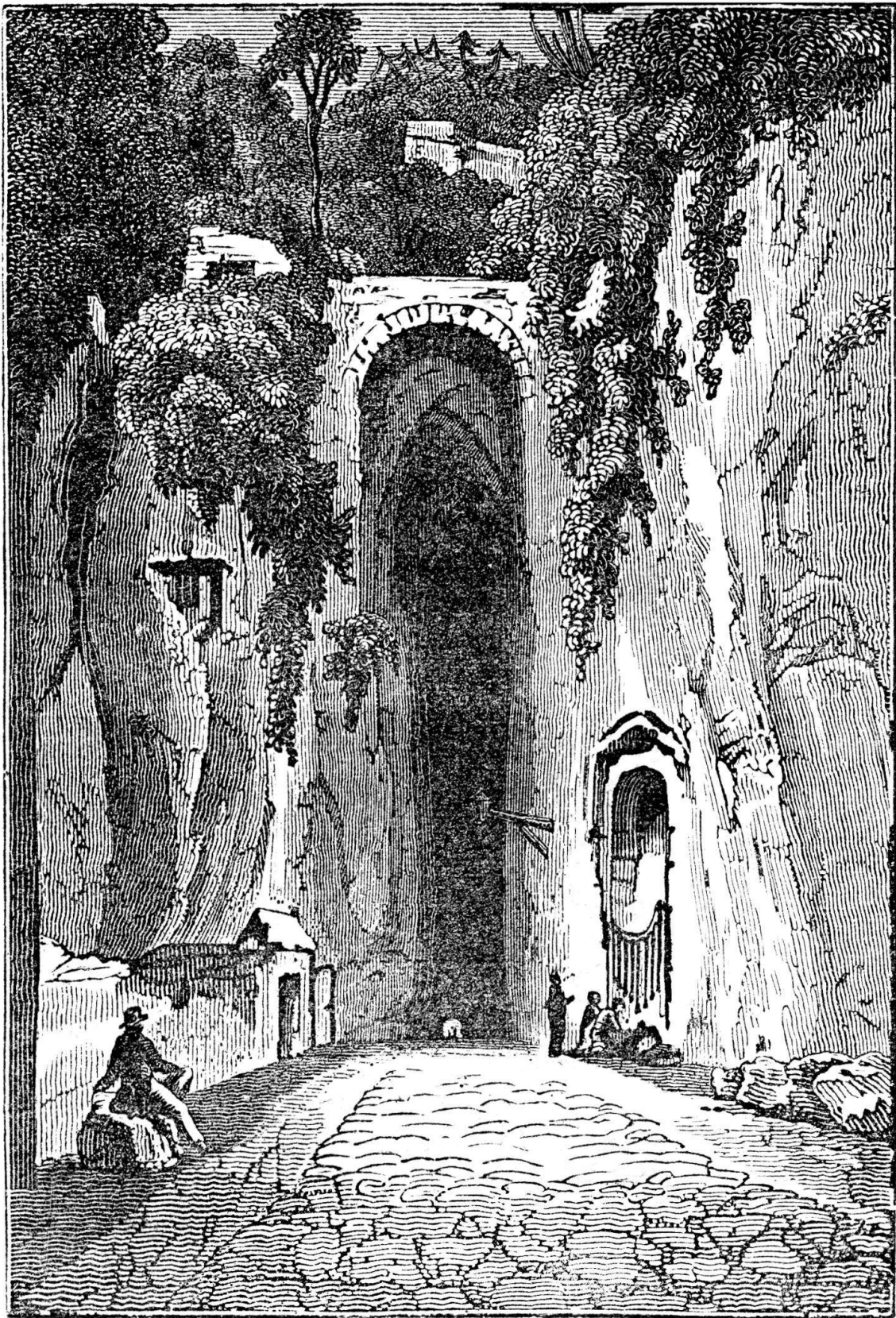
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